

Inkling's Relative Attention

A compact mathematical breakdown

Based on the public Hugging Face blog, model configuration, and Transformers implementation (16 July 2026).

Key idea. For each query token and attention head, Inkling generates a *small vector describing what distances it currently prefers*. That vector selects a scalar positional bias for every query–key pair, and this bias is added directly to the ordinary content-attention logit.

1. Start from ordinary self-attention

Let token i have hidden state $x_i \in \mathbb{R}^D$. For head h , ordinary attention forms

$$q_i^h = W_Q^h x_i, \quad k_j^h = W_K^h x_j, \quad v_j^h = W_V^h x_j,$$

and scores key j for query i using

$$s_{ij}^h = \frac{\langle q_i^h, k_j^h \rangle}{\sqrt{d_h}}.$$

This says whether the *contents* of tokens i and j match, but it does not inherently say whether j is one token ago or one thousand tokens ago. RoPE solves this by rotating q and k . Inkling instead leaves positional information as a separate additive term.

2. The extra projection: a relative feature per token and head

Inkling adds a fourth projection besides Q, K, V :

$$r_i^h = W_R^h x_i \in \mathbb{R}^{d_{\text{rel}}}.$$

For the released model,

$$d_{\text{rel}} = 16.$$

Thus, each query token produces only 16 relative-position coefficients per head. Crucially, r_i^h depends on the token's current hidden state, so positional preference is **content-conditioned**.

3. A learned bank of distance profiles

Each attention layer learns a matrix

$$P \in \mathbb{R}^{d_{\text{rel}} \times E},$$

where E is the maximum distance covered by the explicit relative bias. Row m of P is a learned bias curve over backward distances:

$$P_{m,:} = (P_{m,0}, P_{m,1}, \dots, P_{m,E-1}).$$

The query-specific coefficients r_i^h mix these d_{rel} curves:

$$c_i^h = (r_i^h)^\top P \in \mathbb{R}^E$$

or componentwise,

$$c_i^h[d] = \sum_{m=1}^{d_{\text{rel}}} r_{i,m}^h P_{m,d}.$$

So $c_i^h[d]$ is the positional bias that query i , head h , assigns to a key exactly d tokens behind it.

Key idea. P stores 16 reusable “distance preference curves.” The current query dynamically combines them. Inkling therefore learns neither one universal distance bias nor a huge independent table for every token; it learns a low-dimensional basis plus token-dependent mixing weights.

4. Turn query–key distance into a scalar bias

For query position i and key position j , define the backward distance

$$\delta_{ij} = i - j.$$

Then Inkling gathers the corresponding entry of c_i^h :

$$b_{ij}^h = \begin{cases} c_i^h[\delta_{ij}], & 0 \leq \delta_{ij} < E, \\ 0, & \text{otherwise.} \end{cases}$$

Interpretation:

- $\delta = 0$: the key is the query token itself;
- $\delta = 1$: the key is the immediately preceding token;
- $\delta = 100$: the key is 100 tokens back;
- $\delta < 0$: the key is in the future; the relative term is zero, while the causal mask forbids it;
- $\delta \geq E$: the explicit positional bias is zero. In a global layer, the key can still be selected through content similarity.

5. Final attention equation

Inkling RMS-normalizes q and k per head. Since an RMS-normalized d_h -vector has squared norm on the order of d_h , it uses $1/d_h$ rather than the conventional $1/\sqrt{d_h}$ scale. The logit is

$$z_{ij}^h = \frac{\langle \hat{q}_i^h, \hat{k}_j^h \rangle}{d_h} + b_{ij}^h + \mathcal{M}_{ij}$$

where $\mathcal{M}_{ij} = 0$ for allowed keys and $-\infty$ for forbidden keys. Then

$$a_{ij}^h = \text{softmax}_j(z_{ij}^h), \quad o_i^h = \sum_j a_{ij}^h v_j^h.$$

The positional term is therefore measured in the *same units as an attention logit*. A positive b_{ij}^h multiplies the key's pre-normalization attention weight by $e^{b_{ij}^h}$; a negative value suppresses it.

Long-context log scaling in global layers

For global attention only, Inkling scales both the content and relative terms at late positions:

$$\tau_i = 1 + \alpha \log \left(\max \left(1, \frac{i+1}{n_0} \right) \right),$$

$$z_{ij}^h = \tau_i \left(\frac{\langle \hat{q}_i^h, \hat{k}_j^h \rangle}{d_h} + b_{ij}^h \right) + \mathcal{M}_{ij}.$$

The released configuration uses $n_0 = 128,000$ and $\alpha = 0.1$. Hence $\tau_i = 1$ up to 128k tokens and is about 1.206 near one million tokens. This mildly sharpens the softmax as the number of competing keys grows.

6. Tiny numerical example

Use only two basis curves and four distances:

$$P = \begin{bmatrix} 0.1 & 0.7 & 0.2 & -0.1 \\ 0.5 & 0.2 & -0.3 & -0.4 \end{bmatrix}, \quad r_i^h = \begin{bmatrix} 2 \\ -1 \end{bmatrix}.$$

The query generates

$$c_i^h = (r_i^h)^\top P = 2P_{1,:} - P_{2,:} = [-0.3 \quad 1.2 \quad 0.7 \quad 0.2].$$

Therefore the same query/head adds:

distance d	0	1	2	3
bias $c_i^h[d]$	-0.3	1.2	0.7	0.2

Holding content scores equal, a key one token back receives $e^{1.2-0.2} = e^1 \approx 2.72$ times the unnormalized attention weight of a key three tokens back. Another query token can produce a completely different curve because it has a different r_i^h .

7. Exact tensor view

For batch size B , sequence length L , heads H , relative dimension r , and extent E :

$$X : [B, L, D] \xrightarrow{W_R} R : [B, L, H, r] \xrightarrow{\times P} C : [B, H, L, E].$$

A distance matrix $\Delta_{ij} = i - j$ is used to gather

$$B_{\text{pos}}[b, h, i, j] = C[b, h, i, \Delta_{ij}],$$

for valid distances. Finally,

$$Z = \frac{QK^\top}{d_h} + B_{\text{pos}} + \mathcal{M}.$$

In the Hugging Face implementation, P is shared by the heads of a layer; head-specific behavior comes from the head-specific vectors r_i^h . Every layer has its own W_R and P .

8. Inkling’s released dimensions

Quantity	Global layers	Sliding-window layers
Hidden size D	6144	6144
Attention heads H	64	64
Head dimension d_h	128	128
Relative dimension d_{rel}	16	16
Relative extent E	1024	512
Key/value heads	8	16
Explicitly attendable range	Full context	Last 512 tokens

In a global layer, the explicit relative bias covers distances 0–1023, but content attention can still reach farther keys. In a sliding layer, the extent matches the 512-token attention window.

9. Relative attention versus RoPE

	RoPE	Inkling relative attention
Where position enters	Rotates q_i and k_j before their dot product	Adds a separate scalar b_{ij}^h to the dot product
Dependence on query content	Positional transform is fixed for a position/frequency	Distance curve changes with x_i through r_i^h
Relative information	Emerges inside $q_i^\top k_j$	Explicitly indexed by $i - j$
Interpretability	Distributed across vector rotations	Can inspect a query’s learned curve $c_i^h[d]$ directly
Long distances	Encoded through rotations	Explicit bias becomes zero beyond extent E

10. The mental model to keep

1. The query asks, through Q : *What content am I looking for?*
2. The same query asks, through R : *At what backward distances should I prefer finding it?*
3. $QK^\top$ scores content compatibility.
4. $R^\top P$ produces a dynamic distance preference curve.
5. Adding the two lets attention jointly choose *what* and *how far away*.

Key idea. The whole mechanism is summarized by

$$z_{ij}^h = \underbrace{\frac{\langle \hat{q}_i^h, \hat{k}_j^h \rangle}{d_h}}_{\text{content match}} + \underbrace{\sum_{m=1}^{16} r_{i,m}^h P_{m,i-j}}_{\text{query-conditioned distance preference}} + \underbrace{\mathcal{M}_{ij}}_{\text{causality/window}}.$$

The second term is used only when $0 \leq i - j < E$.

Primary sources

- [Hugging Face blog: Welcome Inkling by Thinking Machines](#)
- [Transformers implementation: modeling_inkling.py](#)
- [Released Inkling config.json](#)